

Module 5 PID Tuning Laboratory

Change one value at a time and select constants using recorded evidence

Student: _____

Date: _____

Pathway: Code Learning / Robot Programming

Team: _____

Predetermined Test Conditions

Target distance or heading: _____ Maximum voltage: _____ Team tolerance: _____

Starting aligner: _____ Surface: _____ Mechanism state: _____

Eight-Trial Tuning Log

Trial	Kp	Ki	Kd	Target	Final error	Over-shoot	Oscillation	Battery	Notes
1									
2									
3									
4									
5									
6									
7									
8									

Controlled Changes

Which variable did you change first, and why?

How did changing K_p affect response speed, final error, overshoot, or oscillation?

How did changing K_d affect overshoot or oscillation?

Integral Investigation

Why does the integral term accumulate?

Why is accumulation limited during voltage saturation?

Why does the template reset accumulated error after crossing the target?

Final Constants

Move Kp: _____ Ki: _____ Kd: _____ Turn Kp: _____ Ki: _____ Kd: _____

Use at least three trials to justify these values.

Ten-Run Completion Check

Trials within tolerance: _____ / 10 Suggested standard: at least 8 / 10

Before-and-after comparison:
